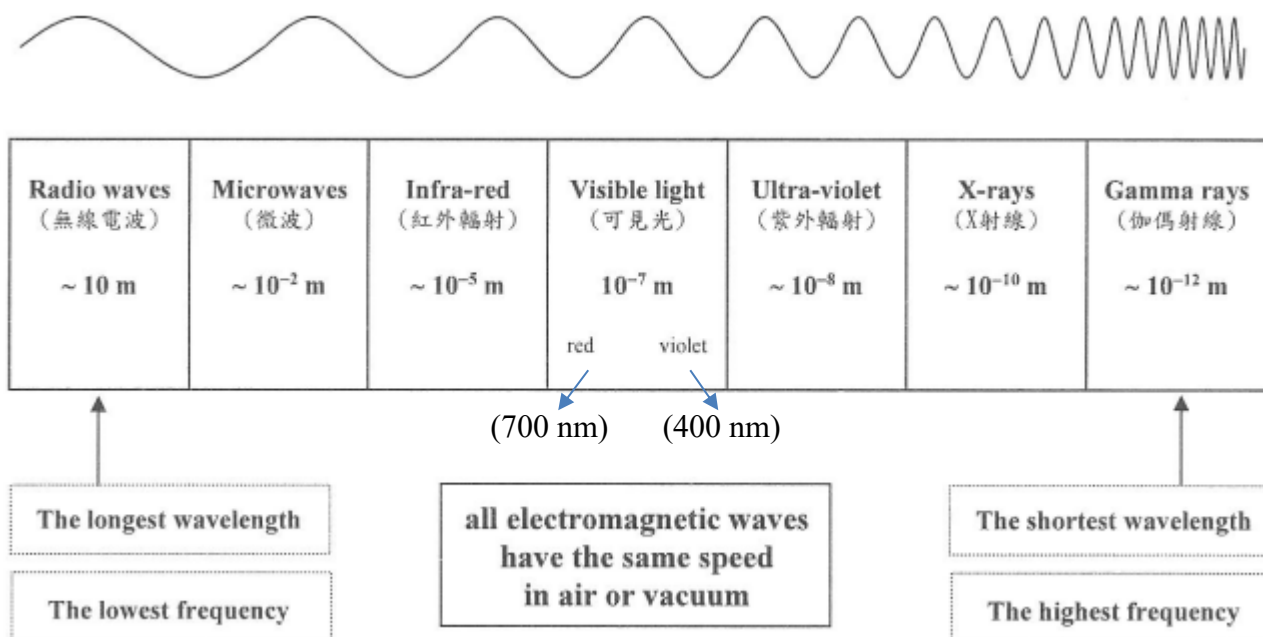


Electromagnetic wave

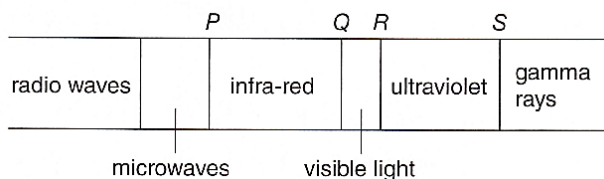


- They are **transverse waves**.
- They **carry energy**.
- They do not require a medium to travel, and **can travel in a vacuum**.
- In a *vacuum*, their **speed c** is fixed at $3 \times 10^8 \text{ m s}^{-1}$. In a material, they travel slower at *various* speeds.
- They obey $v = f\lambda$.
- They show **reflection, refraction, diffraction and interference**.

	Source	Detector	Uses	Wavelength
Radio waves	Radio and TV transmitter	Antenna	● Radio/TV broadcasting ● Mobile phone communication	$10^{-1} - 10^4 \text{ m}$
Microwaves	Microwave transmitter	Microwave receiver	● Microwave oven ● Satellite communication ● Radar	$10^{-3} - 10^{-1} \text{ m}$
Infrared	Objects above 0 K	Skin, phototransistor	● Remote control used in TV ● Auto-focus of camera ● Infra-red thermometer	$10^{-6} - 10^{-3} \text{ m}$
Visible light	Hot objects, Sun	Human eye	●	$4 \times 10^{-7} - 7 \times 10^{-7} \text{ m}$
Ultraviolet	UV lamp, Sun	Fluorescent material	● To sterilize drinking water ● Checking banknotes	$10^{-9} - 10^{-8} \text{ m}$
X-ray	X-ray tube	Photographic film	● Production of see-through images (human body or luggage)	10^{-10} m
Gamma ray	Radioactive substance	GM tube, photographic film	● Medical uses ● Sterilization (killing bacteria in food) ● Radiotherapy	10^{-12} m

1. Student draws the electromagnetic spectrum. However, the part for X-ray is missing. Where should the part for X-ray be?

- A. P
B. Q
C. R
D. S



D

2. Which of the following shows the correct order of five electromagnetic waves arranged in the order of decreasing frequency?

- A X-rays, infra-red, microwaves, radio waves, ultraviolet
B Ultra-violet, infra-red, radio waves, X-rays, microwaves
C Radio waves, microwaves, ultraviolet, infra-red, X-rays
D X-rays, ultraviolet, infra-red, microwaves, radio waves

D

3. A certain medical light has a period T, where $10^{15}T = 1$ s, what types of EM wave does it belong to?

- A Infra-red
B visible light
C ultra-violet
D X-ray

$$\begin{aligned} v &= f\lambda \\ 3 \times 10^8 &= \frac{1}{\frac{1}{10^{15}}} \lambda \\ \lambda &= 3 \times 10^{-7} \text{ m} \\ \therefore \text{It is ultraviolet radiation.} \end{aligned}$$

C

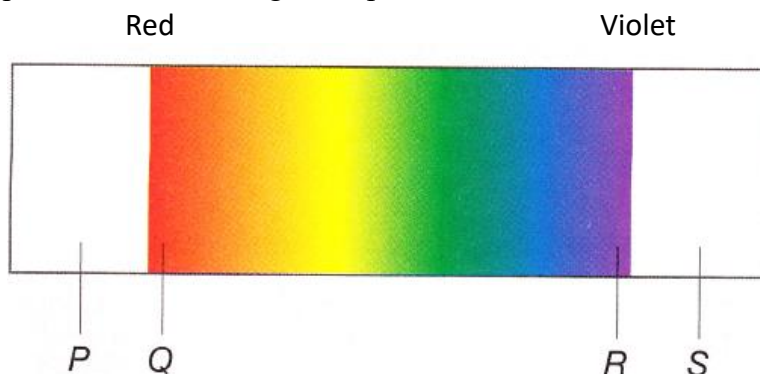
4. Which of the following statements about electromagnetic waves is/are correct?

- (1) The speeds of electromagnetic waves are equal to the speed of light.
(2) All kinds of electromagnetic waves can travel in a vacuum.
(3) Electromagnetic waves are transverse waves.

- A (1) only
B (2) only
C (1) and (3) only
D (1), (2) and (3)

D

5. Figure below shows part of the electromagnetic spectrum.

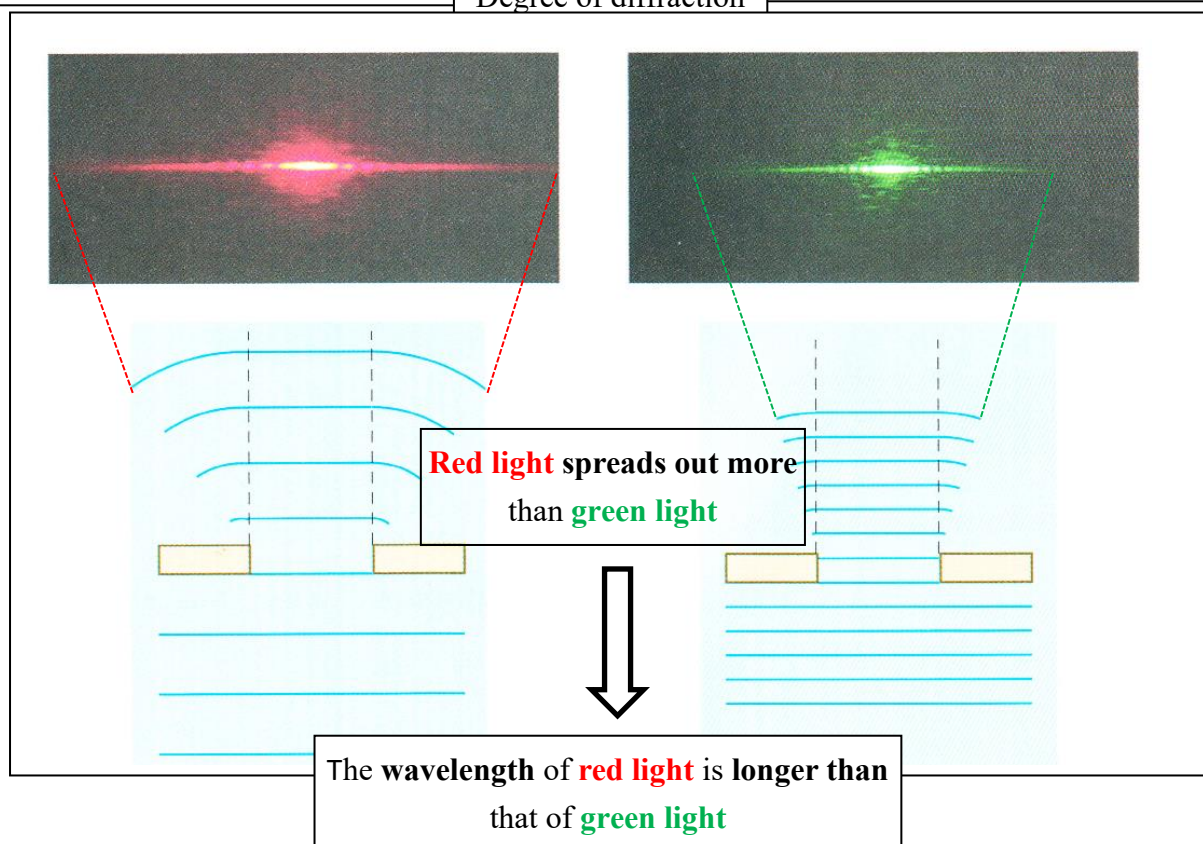
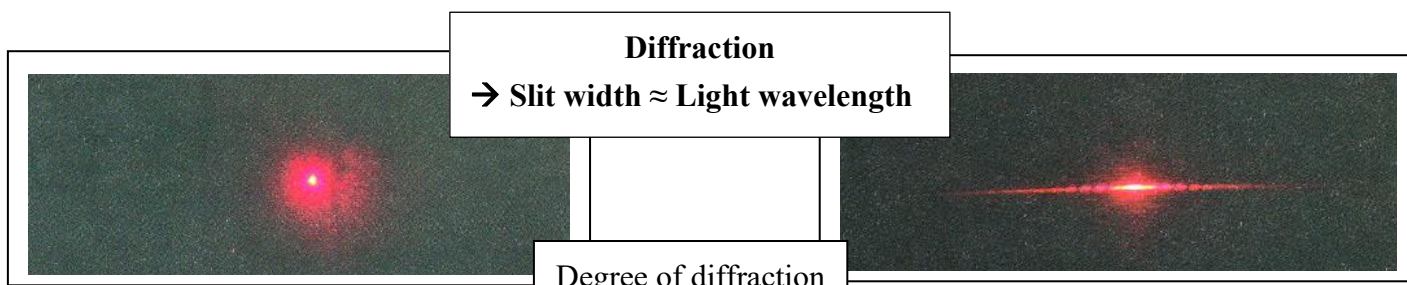
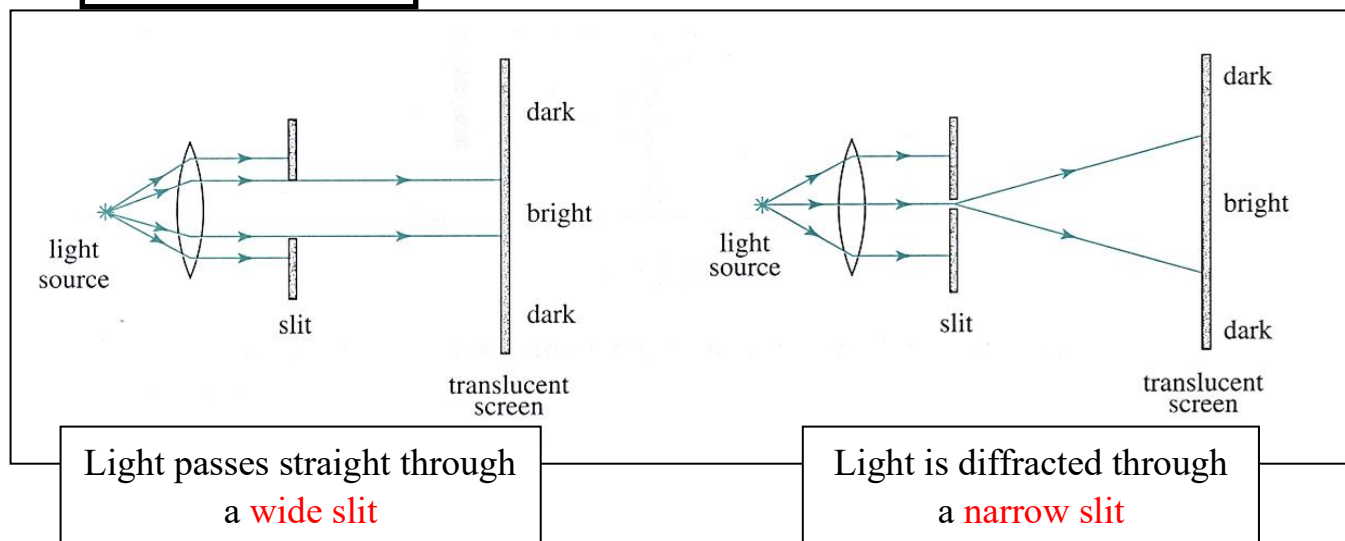


- (a) Which one has the longest wavelength, P, Q, R or S? P
(b) Which one has the highest speed in a vacuum? Same
(c) What is the speed of Q in a vacuum? $3 \times 10^8 \text{ m s}^{-1}$
(d) Name P and S.

P: infra-red radiation

S: ultra-violet radiation

Diffraction of light



6. Which of the following statements concerning visible light and microwaves is/are correct?

- (1) Visible light is electromagnetic wave while microwaves are not.
- (2) Visible light and microwaves travel with the same speed in air.
- (3) Visible light can be diffracted while microwaves cannot.

- A (1) only
- B (2) only
- C (2) and (3) only
- D (1), (2) and (3)

B

7. When a light wave passes through a narrow slit, which of the following wave properties remain(s) unchanged?

- (1) Frequency
- (2) Speed
- (3) Wavelength

- A (1) only
- B (2) only
- C (2) and (3) only
- D (1), (2) and (3)

D

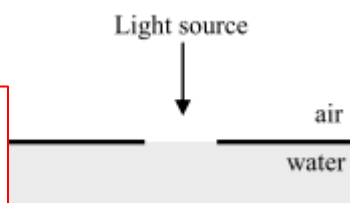
8. 2007 CE MC Q.16

Light travels from air to water through a slit as shown above. Which of the properties of the light in water remain(s) unchanged

- (1) direction of travel
- (2) Speed
- (3) frequency

- A (1) only
- B (3) only
- C (1) and (2) only
- D (2) and (3) only

- | | | |
|-----|---|---|
| (1) | ✗ | Diffraction occurs as the light passes through the narrow slit. The amount of diffraction depends on the size of the slit and the wavelength of the incident wave. The narrower the slit, the more the light will spread out. |
| (2) | ✗ | The speed of a wave is determined by the medium in which the wave is travelling. Since water is optically denser than air, the speed of light decreases as the light travels from air to water. |
| (3) | ✓ | The frequency of a wave depends only on the source. Frequency remains unchanged after reflection, refraction and diffraction. |



B

9. It is difficult to observe diffraction of light in daily life. Which of the following is/are the reason(s)?

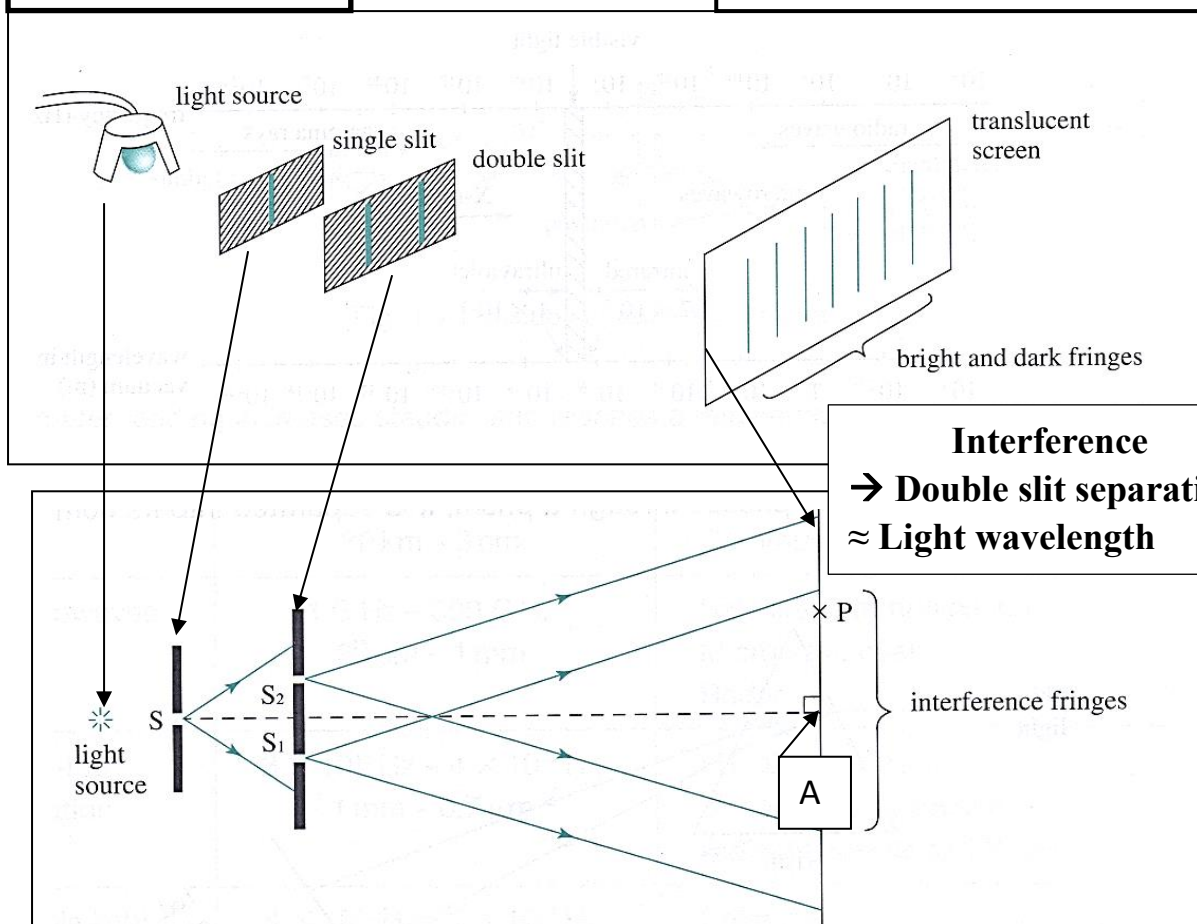
- (1) The wavelength of visible light is very short.
- (2) The light rays that we see in daily life have various wavelengths.
- (3) The speed of light is very high.

- A (1) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

A

Interference of light

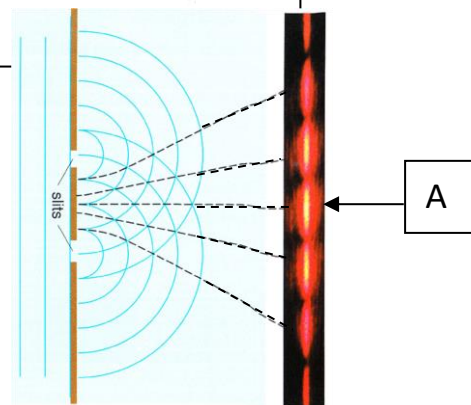
(i) Young's double slit experiment



10. What kind of interference will occur at point A?

Explain your answer.

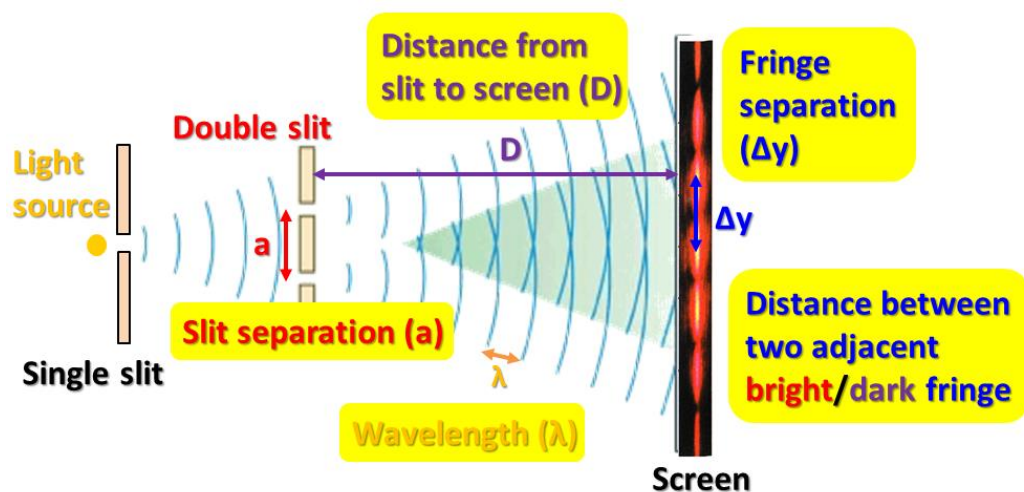
Answer : ∵ A is at equidistant from S₁ and S₂
Path difference = 0 λ
 ∴ **Constructive interference occur at point A**



- Interference occurs on the screen due to different path difference Δ of the two light from the coherent sources S₁ and S₂.
- If path difference at a point is $n\lambda$, constructive interference occurs, a bright fringe is seen.
- If path difference is $(n + \frac{1}{2})\lambda$, destructive interference occurs, a dark fringe is seen.
- Alternative bright and dark fringe can then be observed on the screen.

Remarks:

1. The fringes are **evenly spaced**.
2. D (~1-2 m) should be much longer than a (~0.1-0.5 mm) to achieve small angle since the formula expression is **correct only for small angle** near the central line
3. If the distance D is too large, the bright fringes on the screen would be too **dim** to be observed.



$$\Delta y = \frac{\lambda D}{a}$$

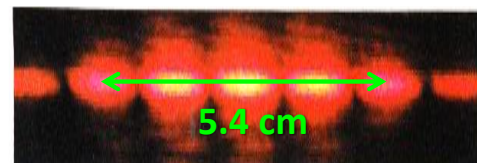
λ, D, a constant
→ Δy constant



Evenly spaced fringes

11. Below is the interference pattern produced by a laser beam. What is the wavelength of the laser?
(double-slit separation = 0.35 mm , distance from double-slit to screen = 7 m)

Ans: 6.75×10^{-7} m



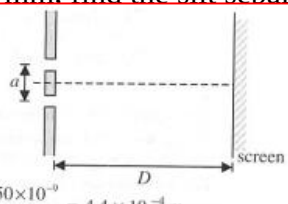
12. In a Young's double slit experiment using monochromatic light of wavelength 550 nm, the fringes are observed through the eyepiece of a travelling microscope at a distance 1.0 m from the double slits. If the separation between the first and the fifth dark fringes is 5.0 mm. find the slit separation

- A 0.3×10^{-4} m
B 1.3×10^{-4} m
C 4.4×10^{-4} m
D 5.5×10^{-4} m

C

There are four fringe separations between the 1st and 5th dark fringes.
 $\therefore$ fringe separation = $s = \frac{5}{4} = 1.25$ mm

By $s = \frac{\lambda D}{a}$,
 $1.25 \times 10^{-3} = \frac{(550 \times 10^{-9}) \times 1}{a}$ $a = \frac{550 \times 10^{-9}}{1.25 \times 10^{-3}} = 4.4 \times 10^{-4}$ m



C

13. In a Young's double slit experiment, the separation between the fringes is increased by increasing
- (1) the distance between the slits and the screen.
 - (2) the slit separation.
 - (3) the wavelength of the light.

- A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

B

14. In a Young's double slit experiment, which of the following combinations of monochromatic light, the slit separation and the slit to screen distance would produce the widest fringe separation on the screen

	Monochromatic light	Slit separation	slit-to-screen distance
A.	red light	1 mm	1 m
B.	red light	1 mm	2 m
C.	red light	2 mm	1 m
D.	green light	1 mm	2 m

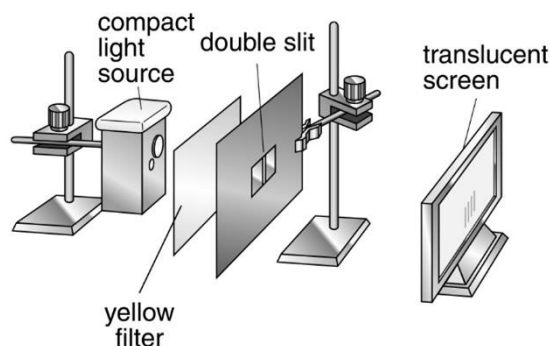
By $s = \frac{\lambda D}{d}$

Monochromatic light : Larger $\lambda \Rightarrow$ red light
 Slit-separation : Smaller $d \Rightarrow$ 1 mm
 Slit-to-screen separation : Larger $D \Rightarrow$ 2 m

B

15. In the following figure, a Young's double-slit experiment is set up and yellow fringes are found on the screen. If the yellow filter is replaced with a red and a blue one, how will the separation between the fringes change as compared with that of the yellow fringes?

	Red filter	Blue filter
A	Increases	Increases
B	Increases	Decreases
C	Decreases	Increases
D	Decreases	Decreases

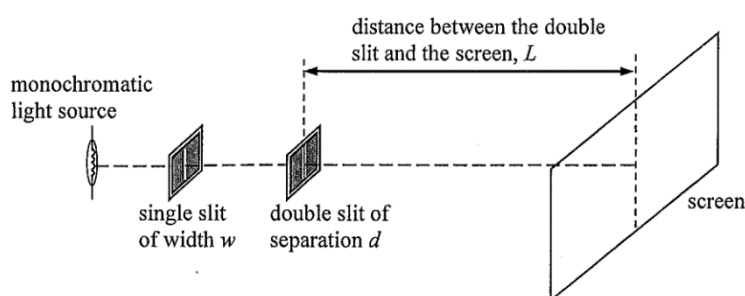


B

16. The figure shows a typical set-up of Young's double slit experiment.

Which combination below is the best setting for displaying an observable fringe pattern on the screen?

	w	d	L
A.	0.1 mm	1 mm	10 m
B.	0.1 mm	1 mm	1 m
C.	1 mm	0.1 mm	1 m
D.	1 mm	0.1 mm	0.1 m



C

By $\Delta y = \frac{\lambda D}{a}$ and assuming the wavelength of the light source is 500 nm, the fringe separation on the screen for A and C are both 5 mm, while that for B and D are both 0.5 mm. Separation of 0.5 mm is too small to be observed, so B and D should be eliminated. As the single slit width w in A is smaller than that in C, so a smaller amount of light is allowed to enter. Also, the screen distance L in A is longer than that in C, so the fringes on the screen in A is less bright. So C is better than A as the fringes would be brighter.

17. In a Young's double slit experiment, the displacement of the 7th order dark fringe is 5 cm from the central bright fringe. Given that the wavelength of the light source is 600 nm and the distance between the slit and the screen is 1 m. Find the slit separation of the double slit.

- A. 0.078 mm
B. 0.084 mm
C. 0.09 mm
D. 0.102 mm

$$d = \frac{D\lambda}{a}$$

$$\frac{0.05}{6.5} = \frac{1 \times 600 \times 10^{-9}}{a}$$

$$a = 7.8 \times 10^{-5} \text{ m or } 0.078 \text{ mm}$$

A

18. Two slits X and Y are illuminated by monochromatic light and the interference pattern is formed on a screen as shown in the figure. The slit separation is 0.15 mm and the distance between the slits and the screen is 1.5 m. If the fringe separation is measured as 6 mm and P is the 10th dark fringe, find the path difference from X to Y to P.

- A. 5 100 nm
B. 5 400 nm
C. 5 700 nm
D. 6 300 nm

$$d = \frac{D\lambda}{a}$$

$$6 \times 10^{-3} = \frac{1.5 \times \lambda}{0.15 \times 10^{-3}}$$

$$\lambda = 6 \times 10^{-7} \text{ m or } 600 \text{ nm}$$

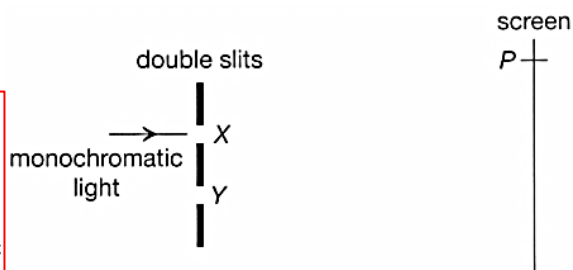
The path difference for dark fringes is given by:

$$d = (m + \frac{1}{2})\lambda$$

For the 10th dark fringe, $m = 9$.

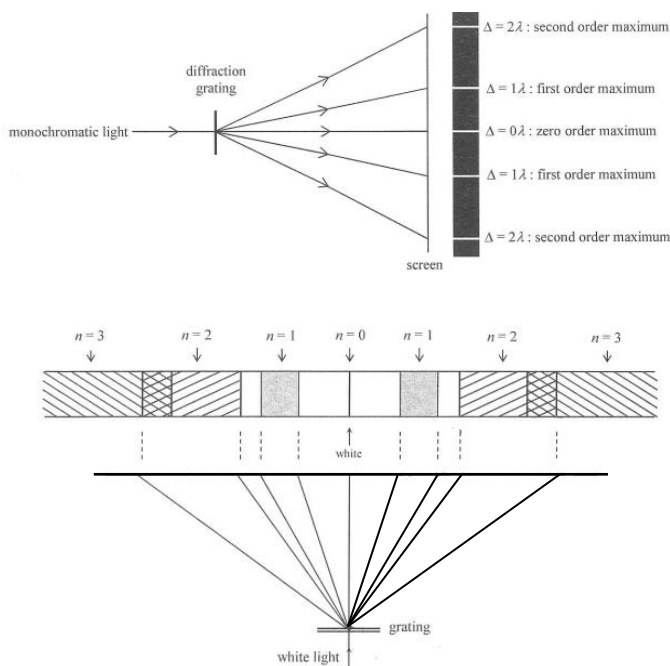
$$d = (9 + \frac{1}{2}) 600$$

$$= 5 700 \text{ nm}$$



C

(ii) Plane transmission grating



$$d \sin \theta = n \lambda$$

Where

d is grating spacing

n is the n th order bright fringe

λ is monochromatic light wavelength

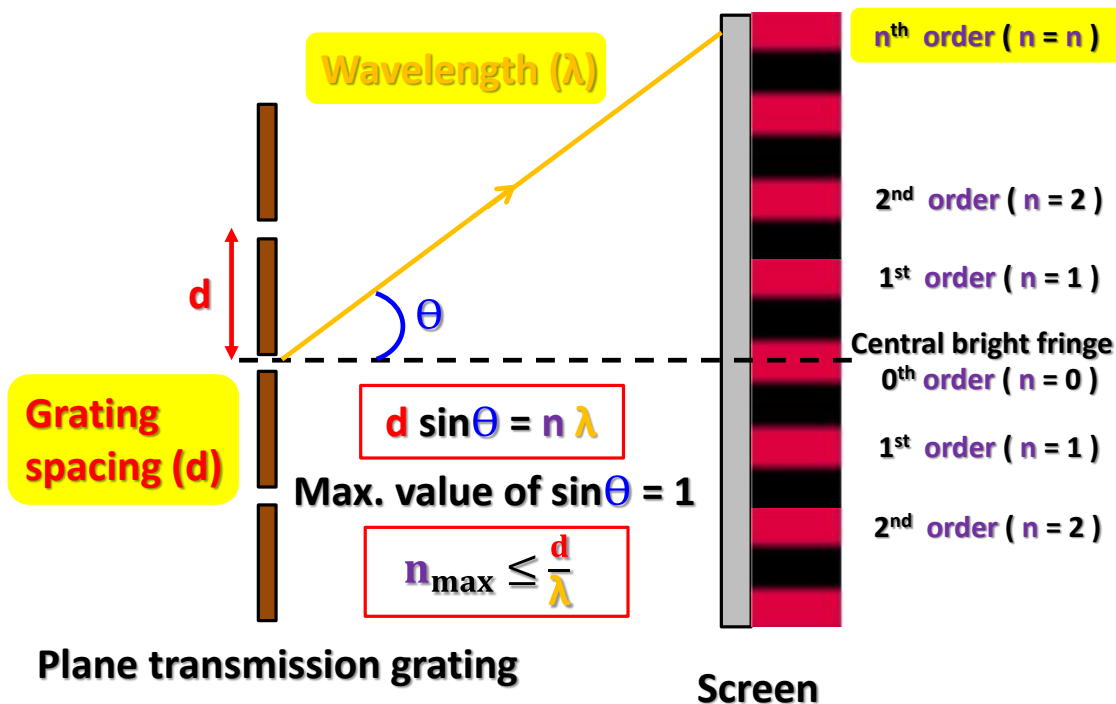
Remarks:

- The maximum diffracted angle is 90°
- Between the bright fringes, the region is all dark.

- The **zeroth order** bright fringe is **white** since every colour of light reach this position.
- For other orders of maxima, continuous spectrum with **red** colour at the **outer** edge and **violet** at the **inner** edge can be seen.
- The higher the order of spectrum (maximum), the wider is the spectrum.
- All the orders of continuous spectrum **higher than the second order** must have **overlapping**.

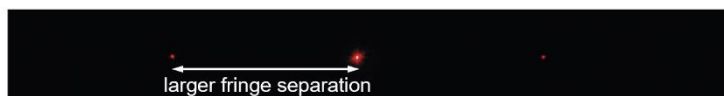
19. If the grating is of 200 lines per mm and red light is used, what is the maximum number of bright fringes ?

Ans: 15

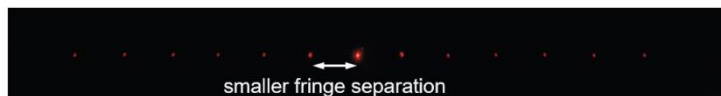


Factors affecting the interference pattern

1. **Grating spacing:** A grating with larger grating spacing produces fringes of small fringe separation.

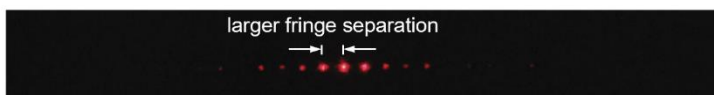


Bright fringes formed by a fine grating (smaller grating spacing)



Bright fringes formed by a coarse grating (larger grating spacing)

2. **Wavelength:** A shorter wavelength of light produces fringes of small fringe separation.

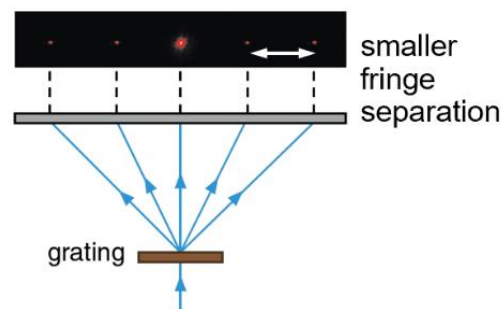
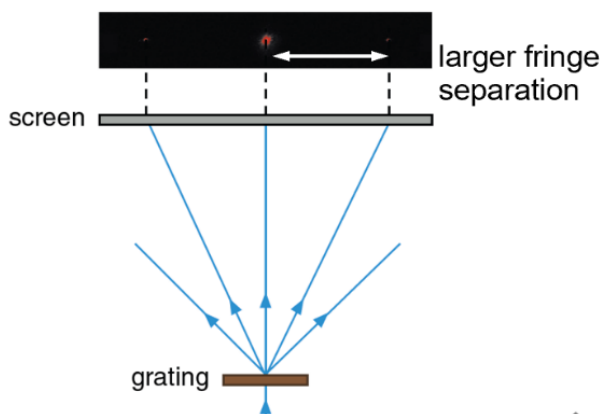


Bright fringes formed by red light (longer λ)

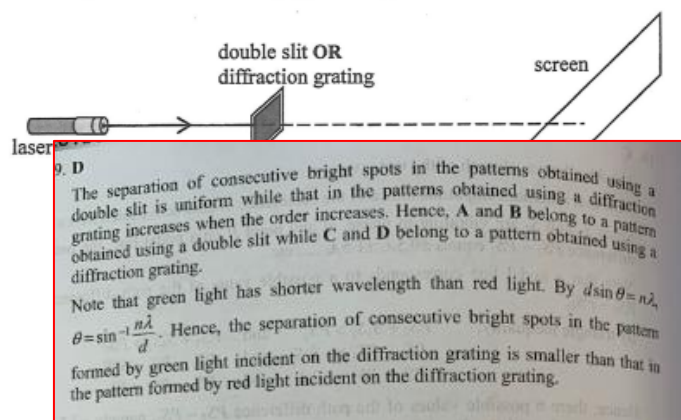
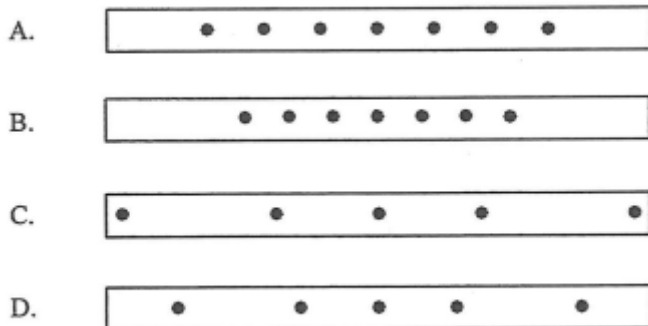


Bright fringes formed by blue (shorter λ)

3. **Grating-screen distance:** A small distance between the screen and the grating produces fringes of small fringe separation



20. A double slit and a diffraction grating are used in turns in the above set-up such that red and green laser lights are directed one after the other on each of them. The four resulting patterns of bright spots obtained on the screen are shown below. Which one belongs to the pattern formed by green light incident on the diffraction grating?



21. When a monochromatic green light of wavelength 550 nm is shone normally on a diffraction grating of 2000 lines per cm, what is the highest order of spectrum observed?

- A. 7
B. 8
C. 9
D. 10

By the diffraction grating equation, $d \sin \theta = n\lambda$
 $\frac{0.01}{2000} \sin 90^\circ = n(550 \times 10^{-9}) \Rightarrow n = 9.09$
Therefore, the highest order is 9.

22. When a monochromatic light of wavelength 4×10^{-5} cm is directed perpendicularly to a plane transmission grating, the second order bright fringe is formed at an angle of 30° to the normal of the grating. What is the number of lines per cm of the grating?

A. 3125
B. 6250
C. 13 000
D. 25 000

B

23. A beam of monochromatic light of wavelength 600 nm is incident normally on diffraction grating having a ruling of 2000 lines per cm. Find the diffraction angle of the third order maximum

A. 17.5°
B. 20.7°
C. 21.1°
D. 28.7°

$$d \sin \theta = m\lambda$$

$$\frac{0.01}{2000} \times \sin \theta = 3 \times 600 \times 10^{-9}$$

$$\therefore \theta = 21.1^\circ$$

C

24. A light beam consisting of wavelengths λ_1 and λ_2 is incident normally on a diffraction grating. The third-order diffraction of wavelength λ_1 coincides with the fourth-order diffraction of wavelength λ_2 in the resulting pattern. If λ_1 is 680 nm, find λ_2 .

A. 510 nm
B. 680 nm
C. 907 nm
D. It cannot be determined because the grating spacing is unknown.

the fourth-order diffraction of wavelength λ_2 , $3\lambda_1 = 4\lambda_2$.
 $\lambda_2 = 3(680) \div 4 = 510$ nm

A

25. A beam of monochromatic light falls at normal incidence on a diffraction grating. Third-order diffracted beams are found at angles of 45° to the incident direction. What is the highest order of diffracted beam that can be observed?

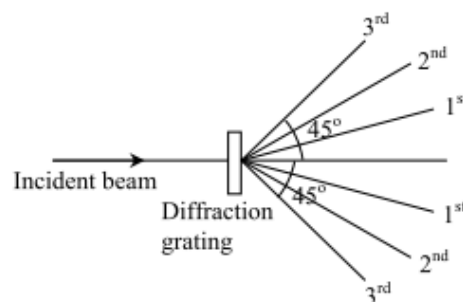
A. 3rd order
B. 4th order
C. 5th order
D. 6th order

① $d \sin 45^\circ = 3\lambda$

② $d \sin 90^\circ = n\lambda$

$\therefore n = 4.24$

Thus the highest order is the 4th order.

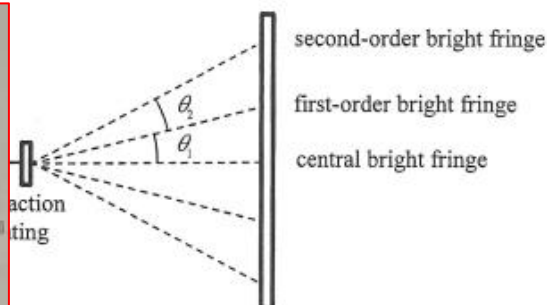


B

26. The figure below shows some of the bright fringes formed when monochromatic light passes through a diffraction grating. Which of the following is/are correct?

- (1) $\theta_1 = \theta_2$
(2) The maximum order of bright fringe is 4 if $\theta_1 = 20^\circ$.
(3) θ_1 will decrease if the experiment is performed in water but not in air.
- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(1) $\times$ Note that $\sin^{-1} \frac{2\lambda}{d} < 2 \sin^{-1} \frac{\lambda}{d}$. Hence, $\theta_1 > \theta_2$.
(2) $\times$ If $\theta_1 = 20^\circ$, Then
 $\frac{\lambda}{d} = \sin 20^\circ$
By $d \sin \theta = n\lambda$, $n = \frac{d \sin \theta}{\lambda} = \frac{\sin \theta}{\sin 20^\circ} \leq \frac{1}{\sin 20^\circ} = 2.92$
 $\therefore$ The maximum order of bright fringe is 2 if $\theta_1 = 20^\circ$.
(3) $\checkmark$ If the experiment is performed in water but not in air, then since λ will decrease, by $\theta_1 = \sin^{-1} \frac{\lambda}{d}$, θ_1 will decrease.



B

27. A plane transmission grating ruled with 5000 lines per cm is illuminated normally by a white light. If the wavelengths for yellow light and violet light are $0.6 \mu\text{m}$ and $0.4 \mu\text{m}$ respectively, which of the following statements is **false**?
- A The central image is white.
B The violet end of the first-order spectrum is closer to the central image than the red end of the first-order spectrum.
C The second-order bright fringe of yellow light coincides with the third order bright fringe of violet light.
D There is no fourth-order bright fringe for violet light.

D

28. A beam of red (R) and violet (V) light strikes normally on a diffraction grating. Several orders of diffraction of the light beam are produced on each side of the central axis. The wavelength of the red light is 700 nm and that of the violet light is 400 nm . How will the second- and third- order diffracted beams of both colours appear? (The central axis is marked with a dot and the diffracted beams of other orders are not shown.)

	Central axis				
A.	•	V	R	V	R
B.	•	V	V	R	R
C.	•	V	R	R	V
D.	•	R	R	V	V

By $\sin \theta = \frac{n\lambda}{d} \propto n\lambda$, $n\lambda \uparrow \Rightarrow \theta \uparrow \Rightarrow$ more apart from central axis

Second-order of violet light : $n\lambda = (2)(400) = 800$
Second-order of red light : $n\lambda = (2)(700) = 1400$
Third-order of violet light : $n\lambda = (3)(400) = 1200$
Third-order of red light : $n\lambda = (3)(700) = 2100$
 $\therefore$ From left to right, the colours are respectively V, V, R, R.

B

29. Light of wavelength 500 nm is incident normally on a plane diffraction grating having a ruling of 5000 lines per cm. Which of the following descriptions is /are correct?

- (1) The spacing between the ruling is 2000 nm
(2) The angle through which the second-order spectrum is diffracted is 30°
(3) No third order spectrum is observable

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

✓	(1)	Grating spacing $d = \frac{1 \times 10^{-2}}{5000} = 2 \times 10^{-6} \text{ m} = 2000 \text{ nm}$
✓	(2)	By $d \sin \theta = n\lambda$ $\therefore (2 \times 10^{-6}) \sin \theta_2 = (2)(500 \times 10^{-9}) \therefore \theta_2 = 30^\circ$
✗	(3)	$\sin \theta_3 = 3 \sin \theta_1 = 1.5 \sin 30^\circ = 1.5 \sin 30^\circ \therefore \theta_3 = 48.6^\circ \therefore$ Third order spectrum exists.

C

30. A student views the straight filament of a ray-box lamp through a double-slit. The lamp is covered by a green filter. How would the interference pattern observed be affected if the following changes are made independently?

- (a) The green filter is replaced with a yellow one.
(b) Another double-slit with a smaller slit separation is used.

The fringes become more widely separated.

The fringes become more widely separated.

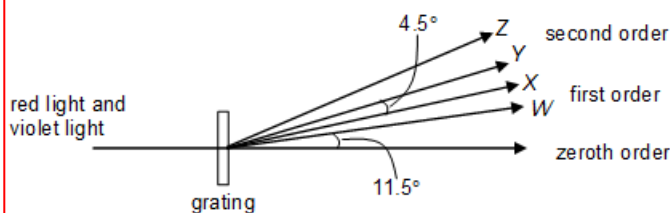
- (c) The experiment is performed under water.

The fringes become closer together.

31. A beam consisting of red and violet lights is shone normally on a plane transmission grating that has ruling of 500 lines per mm. The figure below shows the bright fringes of the first 2 orders.

The angular separation between the first-order fringe at W and the 0th-order fringe is 11.5° . The angular separation between the first-order fringe at X and the second-order fringe at Y is 4.5° .

- What is the colour of the fringe W ? (1 mark)
- Find the wavelength of the violet light. (3 marks)
- Find the wavelength of the red light. (3 marks)
- Find the maximum order of fringe of the red light. (2 marks)



(a) Violet

(b) Grating spacing $= \frac{1 \times 10^{-3}}{500} = 2 \times 10^{-6} \text{ m}$

By $d \sin \theta = n\lambda$,

wavelength $= \frac{d \sin \theta}{n} = \frac{2 \times 10^{-6} \times \sin 11.5^\circ}{1} = 3.99 \times 10^{-7} \text{ m}$

(c) Consider the 2nd order bright fringe of violet light.

$d \sin \theta = n\lambda$

$2 \times 10^{-6} \times \sin \theta = 2 \times 3.99 \times 10^{-7}$

$\theta = 23.5^\circ$

Angle of the 1st order red light $= 23.5^\circ - 4.5^\circ = 19.0^\circ$

Wavelength of red light $= \frac{d \sin \theta}{n} = \frac{2 \times 10^{-6} \times \sin 19.0^\circ}{1} = 6.51 \times 10^{-7} \text{ m}$

(d) $n_{\max} \leq \frac{d}{\lambda} = \frac{2 \times 10^{-6}}{6.51 \times 10^{-7}} = 3.07$

$\therefore$ The maximum order is 3.

32. Monochromatic light of wavelength 620 nm is directed perpendicularly to a plane transmission grating. The first order bright fringe is formed at an angle 20° to the normal of the grating.

- What is the number of lines in 1 cm of the grating? (3 marks)
- What is the maximum number of bright fringes that can be observed? (3 marks)
- Describe and explain the change in the interference pattern formed if monochromatic light of wavelength 400 nm is used instead. (4 marks)

(a) By $d \sin \theta = n\lambda$,

$d = \frac{n\lambda}{\sin \theta} = \frac{1 \times 620 \times 10^{-9}}{\sin 20^\circ} = 1.81 \times 10^{-6} \text{ m}$

Number of lines $= \frac{1}{1.81 \times 10^{-6}} = 552\,000 \text{ lines per m}$
 $= 5520 \text{ lines per cm}$

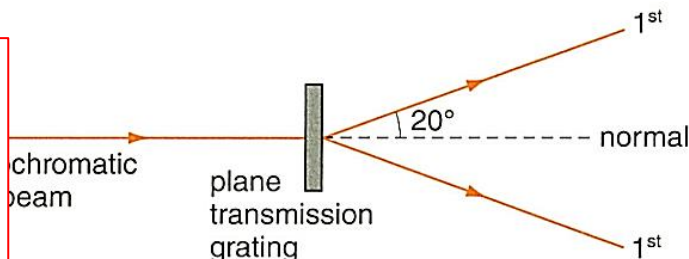
(b) $n_{\max} \leq \frac{d}{\lambda} = \frac{1.81 \times 10^{-6}}{620 \times 10^{-9}} = 2.92$

$\therefore$ The maximum order of bright fringes that can be observed is 2.

Maximum number of bright fringes that can be observed $= 1 + 2 \times 2 = 5$

STRATEGY
Maximum number of bright fringes that can be observed = 1 (the central fringe) + 2 (two fringes for each order) $\times n_{\max}$

MISCONCEPTION
Some students wrongly take $\frac{d \sin 20^\circ}{\lambda}$ as $n_{\max}$.



(c) By $d \sin \theta = n\lambda$, θ decreases when λ decreases.

$\therefore$ The fringe separation decreases when light of wavelength 400 nm is used.

By $n_{\max} \leq \frac{d}{\lambda}$, $n_{\max}$ increases when λ decreases.

$\therefore$ More bright fringes can be observed when light of wavelength 400 nm is used.

STRATEGY
 $n_{\max} \leq \frac{1.81 \times 10^{-6}}{400 \times 10^{-9}} = 4.53$

The maximum number of bright fringes that can be